Ziji Sheng

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Education

Huazhong University of Science and Technology *Sep. 2024 – Jun. 2027 (expected)*
Master's degree in Cyberspace Security, School of Cyber Science and Engineering.
GPA: **3.81/4.00 (top 14%)**.

Wuhan University *Sep. 2020 – Jun. 2024*
Bachelor of Engineering in Cyberspace Security, School of Cyber Science and Engineering.
GPA: **3.62/4.00 (top 30%)**.

Research Interests

Natural language processing, knowledge graphs, runtime security for AI agents, LLM/VLM security and reliable reasoning, and embodied AI safety.

Publications

* Equal contribution. Applicant's name in bold.

1. **Ziji Sheng***, Guiyao Tie*, Weidong Wang*, Jiawen Shi*, Junhao Dong, Xiaoye Qu, Shanshan Ye, Daizong Liu, Dengpan Ye, Pan Zhou, and Jing Zhang.
Object Hallucination Mitigation in Large Vision-Language Models via Self-Vision Dual Masking and Uncertainty-Triggered Assembly.
NeurIPS 2026, Main Track (accepted). Conference listing
   - SIGMA mitigates object hallucination without additional training by separating visual evidence with two masks and intervening during decoding only when uncertainty is high. On LLaVA-1.5, it improves average POPE F1 by 5.51 points over regular decoding.
   - **My contribution:** Designed the full method and all experiments, ran most of the experiments, and wrote most of the manuscript.

2. **Ziji Sheng***, Guiyao Tie*, Weidong Wang, Pan Zhou, and Daizong Liu.
LearnerCoMPASS: Intelligent Tutoring System with Dynamic Cognitive Diagnosis and Multi-Model Path Planning.
ACL 2026, Main Conference (Long Paper), pp. 9016–9042.
   - Combines Graph-RAG, dynamic learner diagnosis, and multi-model path search to plan personalized STEM learning sequences. In the paper's evaluation, hallucination decreased from 31.5% for the base LLM to 4.1% for the full system.
   - **My contribution:** Designed the static and dynamic knowledge graphs, the learning-path generation module, and all experiments, and ran most of the experiments.

3. Lin Lu*, Zhigang Zuo*, **Ziji Sheng***, and Pan Zhou.
Merger-as-a-Stealer: Stealing Targeted PII from Aligned LLMs with Model Merging.
EMNLP 2025, Main Conference, pp. 5806–5825.
   - Shows how merging an aligned LLM with a maliciously fine-tuned model can expose targeted personal information. The attack achieved a 76% exact-match rate for email extraction against DPO-aligned LLaMA-2 in the reported setting.
   - **My contribution:** Co-designed the attack framework, wrote the Methods section, ran the main experiments.

Preprint

Guiyao Tie, Jiawen Shi, Dingjie Song, Yixiao Huang, **Ziji Sheng**, et al.
AutoResearch AI: Towards AI-Powered Research Automation for Scientific Discovery.
Survey preprint, arXiv:2605.23204, 2026.

- Surveys AI-assisted research from literature review and idea generation to experimentation, validation, and writing, organized by levels of autonomy.
- **My contribution:** Collected literature, co-wrote sections, and contributed to discussions on the conclusions and future directions.

Research-Relevant Industry Experience

ByteDance – Feishu (Lark) *May 2026 – Sep. 2026*
LLM / Agent Security Algorithm Intern

- Designed a rule language and evidence model linking sensitive resources, outbound actions, and destinations across user input, tool arguments, and API execution records. Contributed to event and information-flow representations for replayable security analysis.
- Developed a MacBERT intent classification and UIE entity extraction pipeline and helped build datasets with 10,000+ intent examples and 21,000+ extraction examples.

Manuscript Under Review

PersistInject: Persistent Prompt Injection Attacks against Multi-Turn LLM Applications.
Co-first author. *Under review at IEEE Transactions on Information Forensics and Security.*
Studies how malicious instructions embedded in model responses can carry the effect of an initial prompt injection into later turns.

Selected Research and Project Experience

Guardrails and Authorization for Tool-Using Agents *Ongoing*
Investigating guardrails that use user intent and agent execution graphs, authorization models for agent actions, and attacks that induce agents to exceed granted permissions. Currently reviewing the literature and running preliminary experiments.

Uncertainty-Aware On-Policy Distillation for Reliable VLA Models *Ongoing*
Investigating whether hallucinated teacher predictions transfer to student policies and whether grounding errors compound during closed-loop execution.

Temporal RAG and Multi-Agent Security-Alert Analysis *Jul. 2025 – Oct. 2025*
Project Lead. Designed a pipeline for screening security logs and retrieving temporal evidence, and implemented orchestration of specialist agents for alert analysis.

Multi-Agent Adaptive Learning-Path Planning *May 2025 – Oct. 2025*
Team Lead. Led system and algorithm development for a tutor that adapts learning paths to an evolving learner model. This research is represented by LearnerCoMPASS.

Human–AI Collaborative Planning *Nov. 2024 – May 2026*
Student Project Lead, provincial/ministerial-level project. Coordinated task decomposition, technical design, and integration of three modules. The team delivered a prototype for planning under incomplete information and changing constraints.

Knowledge-Graph Construction on Domestic Computing Platforms *Mar. 2023 – Sep. 2023*
Core Contributor. Implemented entity and relation extraction with Paddle UIE and refined annotation guidelines to support knowledge-graph construction.

Selected Honors and Awards

- **National Scholarship**, China, 2026.
- **National First Prize**, 19th Challenge Cup, Special Competition, 2025 (Team Lead).
- **National Grand Prize and Champion**, 18th Challenge Cup, Special Competition, 2023 (Core Contributor).
- **National Second Prize and National Third Prize**, 3rd and 4th China Postgraduate Cybersecurity Innovation Competitions, respectively.
- **National Third Prize**, 19th National College Student Information Security Competition (Project Track).

Granted Invention Patents

Second-listed inventor on all three patents. Patent owner: Huazhong University of Science and Technology. Titles are descriptive English translations.

- **Adaptive Learning-Path Planning Based on Multi-Agent Deep Reasoning.** ZL 2025 1 1052184.0. CN121436143B, granted Sep. 22, 2026.
- **Network Security-Alert Analysis Using Multi-Agent LLMs and Dynamic RAG.** ZL 2026 1 0679185.6. CN122226509B, granted Aug. 14, 2026.
- **Multi-Agent Collaborative Retrieval Based on LLM Causal Verification.** ZL 2026 1 0711361.X. CN122262318B, granted Aug. 14, 2026.

Technical Expertise

Research methods: LLM security evaluation, model merging, hallucination mitigation, Graph-RAG, multi-agent planning, and uncertainty-guided inference.
Tools: PyTorch, Transformers, vLLM, Paddle UIE, LangGraph, Neo4j, Milvus, Redis, Linux.